1. Introduction

Managing inventory efficiently is a critical challenge for small businesses, non-profits, and sustainability-focused organizations. Manual tracking methods often lead to errors, overstocking, and significant waste, especially for perishable goods.

The Green-Tech Inventory Assistant is an AI-powered system designed to simplify inventory management while promoting sustainability. It leverages intelligent analytics to predict usage, prevent over-purchasing, and reduce waste.

2. Problem Statement

Many small organizations face the following challenges:

- Manual inventory tracking is error-prone
- Lack of insights leads to overstocking and waste
- Perishable items expire due to poor planning
- Enterprise solutions are expensive and complex

As a result, organizations incur unnecessary costs and environmental impact.

3. Proposed Solution

The proposed system is a lightweight AI-driven inventory assistant that:

- Tracks inventory usage in real time
- Predicts when items will run out
- Suggests optimal reorder timing
- Identifies waste risks and sustainability improvements

The system is designed to be:

- Simple to use
- Cost-effective
- Scalable
- Sustainability-focused

4. System Architecture

The system follows a client-server architecture with AI integration.

Components:
Frontend: React-based dashboard for visualization
Backend: Node.js + Express for API handling

Database: MongoDB for storing inventory data
AI Layer: Gemini API for predictive insights

5. Technology Stack

Frontend : React, Tailwind CSS
Backend : Node.js, Express
Database : MongoDB
AI Engine : Google Gemini API
HTTP Client : Axios

6. Database Design

🔹 Item Collection : Stores inventory data:
name , quantity, unit, threshold, usageHistory, totalUsed, rushAlert

🔹 Insight Collection : Stores AI-generated insights:
key (unique: "ALL"), data (AI response), generatedAt, urgentCount

🔹 Order Collection : Tracks orders
items, customerName, createdAt

7. API Design

GET /api/items → Fetch inventory
POST /api/orders → Add new order
GET /api/ai/insights → Get AI insights
GET /api/ai/insights?force=true → Force refresh

8. AI Integration & Logic

Input to AI:

- Current stock
- Daily usage (processed from usageHistory)

Output from AI:

- Days left, Reorder suggestion
- Stockout probability
- Consumption trend
- Waste risk
- Sustainability score

9. Important Enhancements:

- Daily usage aggregation for accurate predictions
- Caching system (4 hours) to reduce API calls
- Urgent refresh limit (2 times)
- Fallback logic when AI fails

10. Key Features

- AI-powered inventory insights
- Predictive reorder system
- Urgent refresh functionality
- Waste risk detection
- Sustainability scoring
- Caching for performance optimization

11. Challenges & Solutions

Challenges followed by solutions are written below:

- AI giving inconsistent results : Added rule-based constraints
- Incorrect averages : Aggregated daily usage
- Duplicate DB entries : Used upsert instead of create
- High API usage : Implemented caching
- UI errors (map issue) : Fixed data structure consistency

12. Sustainability Impact

The system contributes to sustainability by:

- Reducing overstocking
- Minimizing expired goods
- Encouraging efficient resource usage
- Providing waste reduction insights

13. Future Enhancements

- Visual inventory scanning using AI
- Real-time notifications
- Carbon footprint tracking
- Supplier recommendations
- Advanced analytics dashboard

14. Success Metrics

Waste Reduction: Lower expired inventory

Ease of Use: Faster than spreadsheets
AI Effectiveness: Accurate predictions with fallback

15. Conclusion

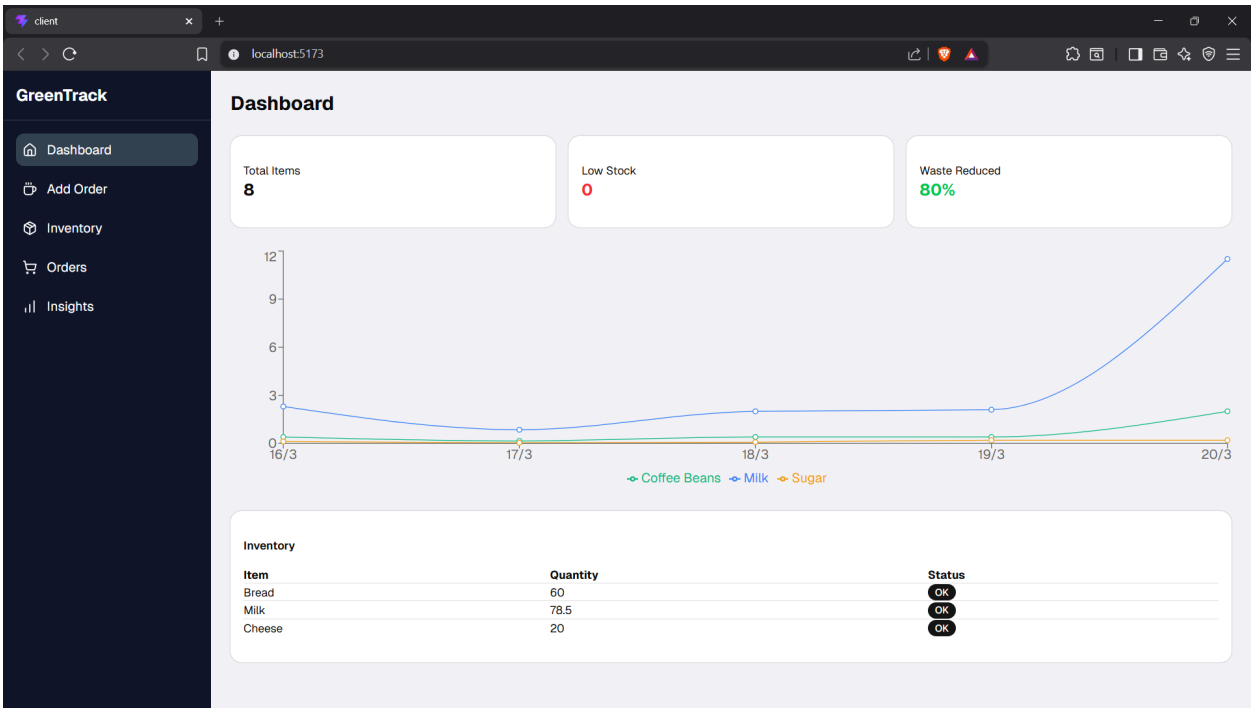
The Green-Tech Inventory Assistant provides an intelligent and accessible solution for inventory management. By combining AI with efficient system design, it helps organizations reduce waste, optimize stock levels, and operate more sustainably.

This project demonstrates how AI can be practically applied to solve real-world problems in a scalable and cost-effective manner.

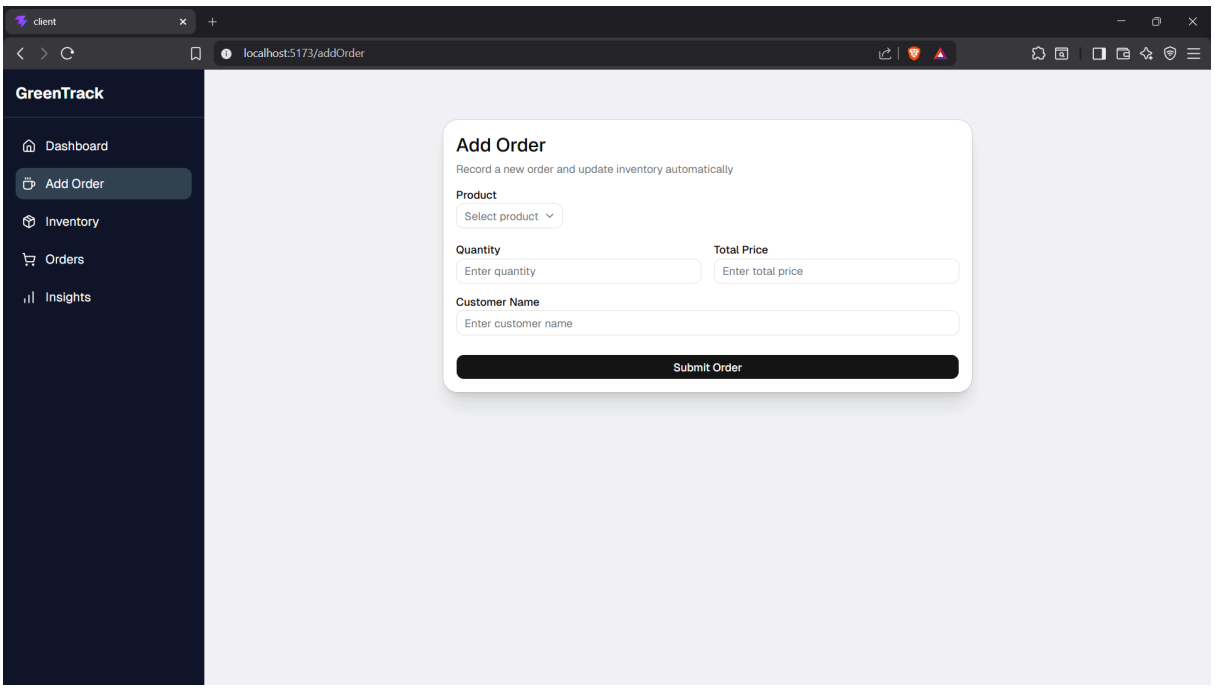
It bridges the gap between traditional inventory systems and intelligent, data-driven decision making.

16. Screenshots

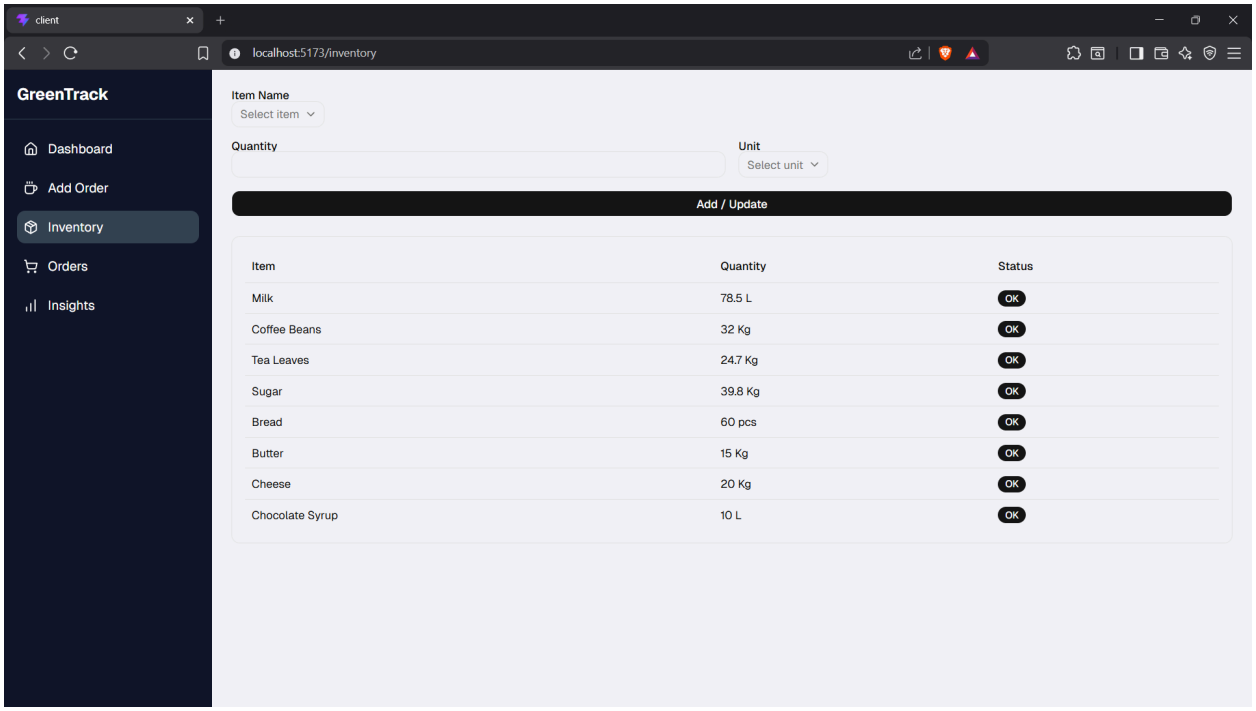
(i) Dashboard Page



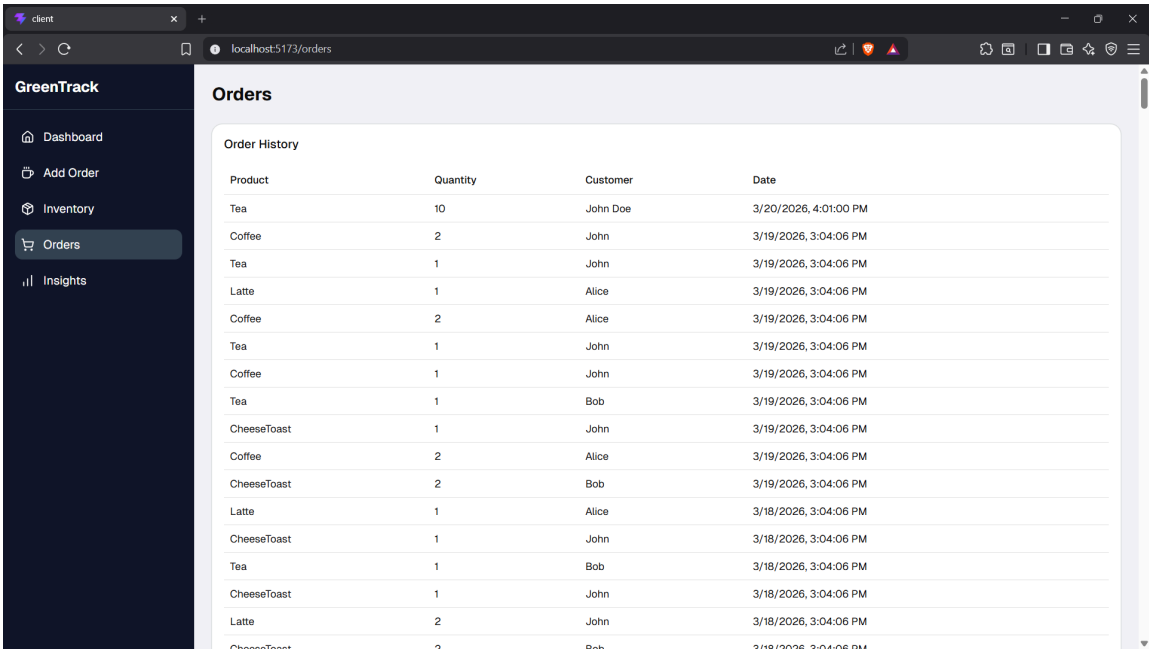
(ii) Add Order Page



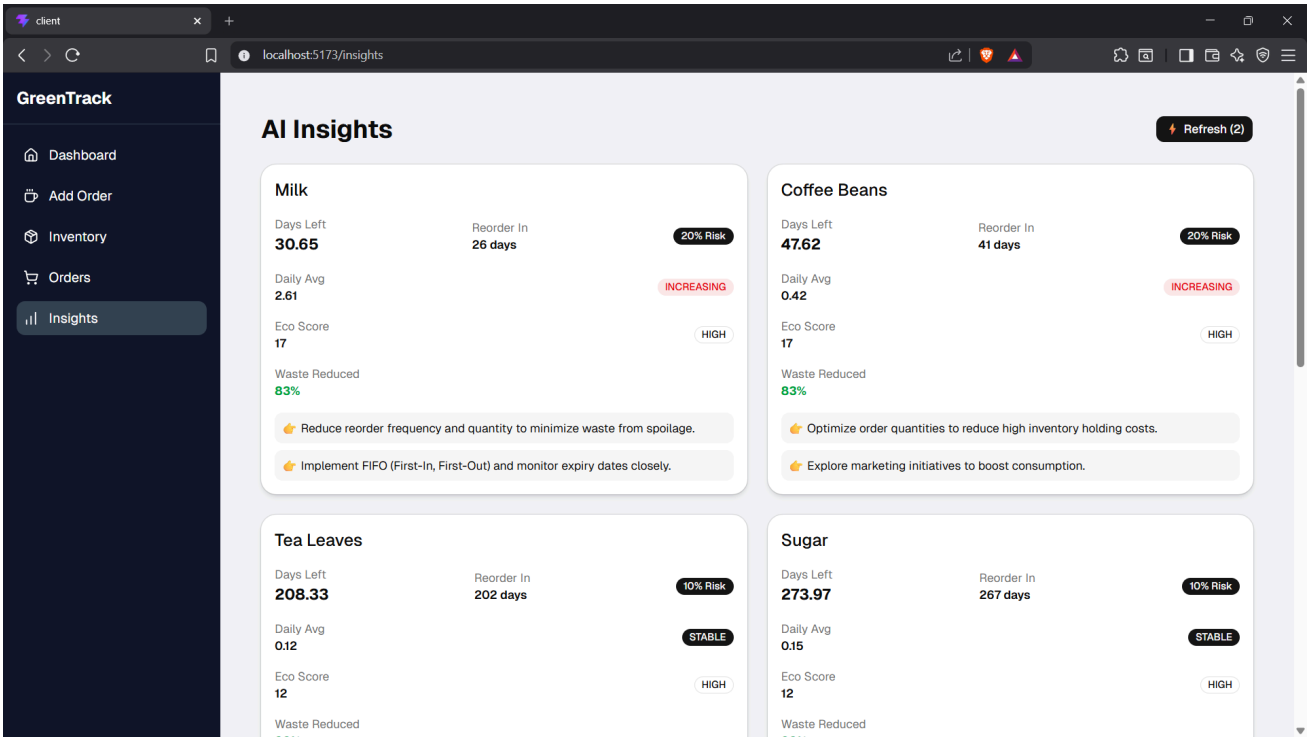
(iii) Inventory Page



(iv) Orders Page



(v) Insights Page



client

localhost:5173/insights

GreenTrack

- Dashboard
- Add Order
- Inventory
- Orders
- Insights

Bread

Days Left
4.79

Reorder In
1 days

70% Risk

Daily Avg
12.52

STABLE

Eco Score
65

LOW

Waste Reduced
35%

👉 Place an urgent reorder to maintain continuous supply.

👉 Consider a small safety stock buffer for this item.

Butter

Days Left
45.45

Reorder In
41 days

20% Risk

Daily Avg
0.33

INCREASING

Eco Score
17

HIGH

Waste Reduced
83%

👉 Reduce reorder frequency and quantity to minimize waste from spoilage.

👉 Implement FIFO (First-In, First-Out) and monitor expiry dates closely.

Cheese

Days Left
21.74

Reorder In
17 days

20% Risk

Daily Avg
0.92

INCREASING

Eco Score
17

HIGH

Waste Reduced
83%

👉 Reduce reorder frequency and quantity to minimize waste from spoilage.

👉 Implement FIFO (First-In, First-Out) and monitor expiry dates closely.

Chocolate Syrup

Days Left
76.92

Reorder In
70 days

0% Risk

Daily Avg
0.13

DECREASING

Eco Score
7

HIGH

Waste Reduced
93%

👉 Reduce future order quantities to prevent overstocking.

👉 Investigate reasons for declining usage and adjust strategy.